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Robot Vandalism: A Senseless Act?

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Abstract

In Human–Robot Interaction (HRI), vandalization of robots is predominantly framed as senseless or immoral behaviors, with mitigation efforts focused on system design improvements or user education. Drawing on sociological theories of vandalism, we examine three cases of robot destruction in public spaces: the burning of Waymo vehicles in Los Angeles (2025), the vandalism of a Knightscope K5 robot in San Francisco (2017), and the destruction of the hitchhiking robot hitchBOT in Philadelphia (2015). We argue that not all acts of robot vandalism are instances of "malicious" destruction. Rather, some can be understood as ideological or political vandalism – expressive or strategic acts embedded within broader social and political struggles. By situating these events in their urban and discursive contexts, our analysis moves beyond explanations grounded solely in individual psychology or system design, and invites a broader reflection within HRI on how robots as socio-technical artifacts become implicated in the politics of public space, power, and collective life.

CCS Concepts

• **Human-centered computing → HCI theory, concepts and models**.

Keywords

robot vandalism, ideological vandalism, robots as socio-technical artifacts, digital ethnography, robots in urban spaces

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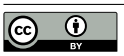
1 Introduction

As more robots become a common presence in public spaces, Human–Robot Interaction (HRI) scholars are paying greater attention to how people, who are not direct users of these systems, interact with them in situ. One type of human behavior that has been under focus of investigation is vandalism and bullying of robots. In HRI literature, robot vandalism, commonly framed as senseless, malicious, or even immoral and devoid of empathy, is predominantly treated as individual outbursts and, as such, as edge cases or operational hazards rather than phenomena warranting deeper social analysis. These framings, while pragmatically useful for addressing immediate operational concerns and potential harm to the robot and other humans [41], tend to overlook the socio-political and cultural dimensions of robot vandalism.

By contrast, social science research on urban vandalism has long recognized that destructive behaviors directed at material property and urban infrastructure are diverse and patterned, with many of such acts reflecting underlying ideologies and social tensions. Within this body of work, Cohen's typology of vandalism, distinguishing, among other forms, ideological (expressing political or social beliefs) and tactical (used instrumentally to achieve a goal) vandalism [10], remains highly influential, offering a structured lens to analyze why people engage in destructive behaviors in urban settings. Taking inspiration from Cohen's work and broader sociological scholarship on vandalism, in this paper, we ask: To which extent and in which way can some acts of robot vandalism be considered ideological? And, if they can, what implications might this have for HRI scholars and robot designers?

To address these questions, we examined three cases of damage or destruction of robots in public spaces: the burning of Waymo vehicles during protests in Los Angeles (June 2025), the vandalism of a Knightscope K5 security robot in San Francisco (December 2017), and the earlier destruction of the hitchBOT robot in Philadelphia (2015). We selected the first two cases based on our (informal) hypothesis that they may carry ideological or political underpinnings. We revisit the hitchBOT case precisely because it has been widely portrayed, both in news media and HRI research, as an instance of "senseless", empathy-devoid vandalism. By re-examining this case through a critical theoretical lens, we aim to explore whether alternative interpretations or overlooked socio-political dimensions might emerge.

Methodologically, our study is positioned within the umbrella of digital ethnography [24]. We first collected online traces of the three events of vandalism, including news reports, social media posts, and public commentary. To interpret this material, we draw on Critical Discourse Analysis (CDA) [15], which links language use to broader social practices; and on frame analysis, an approach to understanding how people organize and interpret experience



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through socially shared frameworks that define "what is going on" in any situation [19].

Importantly, in this paper, our goal is not to suggest that ideological robot vandalism *is* or *is not* a viable (or ethical) strategy for achieving political aims. Addressing this would require a far more extensive empirical and theoretical analysis, firmly rooted within social science debates on protest, social movements, resistance, and political violence – an undertaking that lies beyond the scope of this work. Our aim here is more modest yet, we hope, generative: To invite critical reflection on what current epistemic and methodological practices in HRI may overlook when treating acts of robot destruction as senseless, isolated, and apolitical phenomena. By re-situating these events within their broader socio-political contexts and narratives, we make tentative suggestions for how existing approaches to vandalism in HRI might be expanded to better account for the entanglements between technology, power, and public life.

2 Aggression toward Robots in HRI and Social Robotics

As robots move from controlled environments into public life, researchers have become concerned with how humans respond to their presence, including instances of aggression or harm. Different terms have emerged to refer to destructive behaviors, including abuse, bullying, vandalism, and griefing [7, 39, 45, 48] and even hate crime [41]. Although some scholars have already pointed out that these terms carry distinct meanings [39, 48], much of the ongoing HRI research subsumes a wide range of acts that damage or sabotage a robot's operations under the category of *abuse*. Brščić et al. [7, p.61] define robot abuse as "persistent offensive actions, either verbal or nonverbal, or physical violence that violate the robot role or its human-like (or animal-like) nature."

As can be seen from these definitions, the very use of the term abuse carries connotations of harm directed at quasi-social actors and potential moral patients [3]. And indeed, much of the existing research on destructive behavior toward robots therefore focuses on social robots. As a result of this anthropomorphic orientation, study designs and research questions tend to center on the individual traits – either human or robotic – that underpin abusive behavior, and how these might be modified or mitigated through design [30]), educational strategies [48]), or policy interventions [41]. In other words, abuse is seen as something that is governed primarily by psychological processes [29]; it is individualized and interpreted as either a deviant user behavior or a byproduct of robot features/affordances. [1]

The dominance of individualized and psychologized framings of robot vandalism has methodological consequences. Existing studies often rely on controlled experiments or in-the-wild observations in semi-controlled environments (e.g., shopping malls, schools). When motivations for aggressive behaviors are examined, it is most often with children, with aggression being reinterpreted as play [40]. Adults' motivations, especially situated in labor struggles, surveillance resistance, or anti-tech activism, are rarely considered within the scope of HRI.

[1] One notable exception is the work by Luria et al. who conceptualize robot destruction as a non-utilitarian interaction that can carry creative, cathartic, and emotionally meaningful value [36].

This dominant framing of destructive acts, in our view, obscures the fundamentally socio-technical character of robots. As a result, two important dimensions remain overlooked. First, robots (even in the case of social robots) may be targeted not as anthropomorphic actors but as forms of property or infrastructure, whose destruction carries meanings independent of the immediate level of interaction. Second, acts of robot destruction may not simply express antisocial impulses but can manifest broader forms of social or political contestation.

We recognize that the dominance of anthropomorphism and abuse framings in HRI may reflect the relative novelty of service robots in public settings. As these robots proliferate in urban environments, new opportunities emerge to study HRI as embedded in socio-cultural and political contexts. Recent work has begun to foreground the socio-technical dimensions of situated interactions with public-space robots (e.g., [8, 14, 34, 43]). Our work extends these efforts by drawing on social science theories and methods to examine antagonistic behaviors toward robots within the broader social, economic, and political structures in which they are embedded.

3 Vandalism in Social Sciences

Vandalism, as a form of property or infrastructure destruction or damage, has long been a topic of theoretical and empirical inquiry in social sciences, criminology and urban studies. Among the foundational works on the topic remains Cohen's typology of six types of vandalism, differentiating between: acquisitive (property damage committed incidentally in the pursuit of material gain), tactical (acts undertaken to achieve a practical end, such as seeking arrest for food and shelter), vindictive (revenge-motivated destruction), malicious (anti-social damage without specific personal target), play (often by children, motivated by exploration or amusement), and ideological (destruction committed in the name of a wider belief system or cause) [10]. Cohen's proposal was motivated by a critique of how vandalism was commonly understood in public discourse and academic literature as a homogeneous and "senseless" act, often stereotypically associated with working-class youth or "hooligans." Against this reductionist understanding, Cohen argued that vandalism is not a single phenomenon but a patterned form of rule-breaking behavior that can take diverse forms, each meaningful in its own right (ibid.).

Of these six forms, ideological vandalism is an anchoring framing for this paper. Cohen substantiates that ideological vandalism is a form of property destruction where the person engaged in a vandalist behavior has a claim to be acting in accordance with an – either implicitly held, or explicitly stated – set of beliefs; and as acts where the very rule being broken is contested (ibid.). Ideological vandalism thus foregrounds how property damage and destruction may be expressive of (contested) ideological assumptions regarding the socio-economic and socio-political relations that constitute property itself. As such, the act of ideological vandalism ceases to be merely a destructive act, and instead amounts to a social act with a particular socio-economic meaning, albeit one that is itself open to different interpretations depending on the standpoint of the actor and/or the observer.

Later scholarship has refined this understanding. For example, Long and Burke [35] argued that it is more accurate to speak of

political rather than ideological vandalism, as the defining feature of such acts lies not only in belief (i.e., ideologies as cognitive belief systems which do not necessarily translate into action) but *action* understood here as social practices of physical and material destruction (ibid.). According to them, the actions of political vandalism are designed, first and foremost, to exert pressure on those who may be in some way associated with the "target"; or, alternatively, to attract attention of wider audiences, galvanize supporters, and invite further participation.

Importantly, vandalism is rarely – if ever – a neutral and objective category. Instead, what we classify as vandalism, and the strategies deployed to respond to it, are in themselves political. In the words of Long and Burke, "What passes for vandalism, it was suggested, depends very much on the (often politically motivated) eye of the beholder" [35, p.154]. That is, the same act may be condemned as criminal damage by opponents, yet framed as justified by supporters of the cause.

4 Destruction of Robots in Public Spaces: Three Case Studies

We now examine three cases of robot vandalism in public spaces: the burning of Waymo vehicles in Los Angeles (2025), the vandalism of a Knightscope K5 security robot in San Francisco (2017), and the destruction of the hitchBOT robot in Philadelphia (2015). We selected the first two cases because they represent moments when robots in public spaces became entangled in broader social and political tensions, providing a productive ground for exploring the extent to which, and in what way, any of these instances can be considered ideological vandalism. As a contrasting example, we revisit the widely discussed hitchBOT case – an emblematic example in HRI often framed as a senseless act of destruction – to explore what new insights a different theoretical lens might reveal.

4.1 Methodological Positioning, Data Collection and Analysis

Rooting in the Science and Technology Studies (STS) tradition of foregrounding technologies as situated sociote-chnical artifacts [57, 64], we adopted a digital ethnographic case study approach [24]. For each case study, we collected media reports, social media discussions, and publicly available videos as sources that documented multiple actors' interpretations of the events.

Waymo case study: We collected 111 sources, including 47 media articles spanning a broad political spectrum; 2 Reddit threads with associated comments from different subReddits; 20 TikTok videos with comments; and 19 YouTube videos with comments sorted by popularity. Sources were located via Google searches combining the keywords "Waymo," "fire," "protest," and "Los Angeles," using page-specific operators (e.g., site:theguardian.com, site:Reddit.com). Each search yielded multiple pages of results, which we manually screened by headline and content for relevance; for social media, we prioritized entries with multiple and diverse commentaries. While we focused on the 2025 Los Angeles event, sources referring to earlier Waymo incidents in California were also included for context.

Knightscope case study: We collected 9 media articles spanning a broad political spectrum; 3 Twitter/X posts with large numbers of comments; and 6 YouTube videos with comments, totaling around 50 online comments. Sources were collected via Google searches combining the keywords "SPCA," "robot," and "2017."

hitchBOT case study: We collected 48 sources, including 36 media articles (spanning a broad political spectrum and Philadelphia agencies for local perspective); 6 Reddit threads with comments; 4 Facebook posts with the ten most-liked comments each; and 2 Twitter/X posts with the ten most-liked replies, including hitchBOT's official accounts. Materials were located via targeted Google searches combining the keywords "hitchBOT," "robot," and "Philadelphia," and manually screened by headline and content. We prioritized coverage of hitchBOT's 2015 destruction in Philadelphia, while skimming earlier Canadian and European reports to contextualize the case study.

Our analytic approach combined elements of critical discourse analysis (CDA) [15] and frame analysis [19]. Our aim in adopting the CDA method was to make explicit the relationship between (competing) interpretations of vandalism and the broader social relationships and practices within which these interpretations were formed. Further, as 'CDA has emancipatory objectives' [15, p.125], we also sought to delineate the way that different interpretations uphold or challenge the unequal social relationships from which they emerge, thereby creating the possibility that research contributes to the contestation of relations of domination. Frame analysis, in turn, allowed us to explore how events, actors and robots as socio-technical artifacts were positioned within competing interpretative schemas used to make sense of the events of vandalism. More concretely, this synthetic approach to analysis allowed us to gain greater clarity regarding the different ways in which acts of vandalism are understood, and offer explanations rooted in broader social contexts for why particular understandings or interpretations are adopted by different observers – including why there might be competing interpretations, and contestation over those different interpretations.

The analysis was carried out as follows. First, we individually familiarized ourselves with the collected materials, noting recurrent themes, key moments of contestation, and initial impressions of discursive patterns. We then collectively discussed these observations, reflecting on the central concerns of each case study and refining our interpretative focus. Next, we selected a small subset of sources for close analysis, guided by the following questions: i) How are robots discursively framed? ii) How are the people committing acts of destruction framed? iii) Which ideologies or belief systems underpin these framings? Beyond these questions, we chose to include in the analysis the meanings that participants and observers attributed to acts of vandalism, and how these meanings were subsequently taken up, reframed, or contested in public discourse. Once these analytical orientations were established, the first three authors conducted in-depth analyses of individual cases, with each author taking primary responsibility for one case study.

Ethical Considerations. Collecting and analyzing online data presents ethical challenges for digital ethnography, particularly when participants are not informed of data collection [24]. In consultation with the Research Ethics Committee at the first and third

authors' university, we adopted the following approach. For news media and posts by public figures, we cite the outlet and author(s). For social media content by non-public figures, we omit user handles and other identifying information. When quoting individuals, we preserve key phrasing while paraphrasing surrounding text to reduce traceability, and when citing social media commentary, we identify the platform but not the specific thread. Finally, we limited manual data collection to a maximum of 1,000 comments per platform, avoiding automated scraping or bulk harvesting. This approach allowed us to balance the need for rich qualitative analysis with ethical guidelines and platform policies.

5 Outcomes Empirical Analysis

In the following sections, we present the outcomes of our analysis for each case study. Each begins with a brief contextual overview to situate the acts of vandalism, followed by a discussion of discursive framings and ideological underpinnings of these acts based on our analysis.

5.1 Case Study 1: Burning Waymo taxis in L.A.

The vandalization of Waymo vehicles in downtown Los Angeles occurred on Sunday, June 8, 2025, alongside community protests sparked by federal immigration enforcement operations initiated by the Trump administration. To understand the vandalism and its potential ideological dimensions, it is first necessary to situate it within the broader context of these protests. In turn, understanding the protests requires attention to Los Angeles's jurisdictional and political landscape.

Los Angeles is a sanctuary city within California's sanctuary state framework (Senate Bill 54, 2017; [32]). Sanctuary practices emerged in the 1980s as religious and humanitarian responses to Central American refugees and later expanded through shifts in federal immigration policy [13]. These provisions limit local cooperation with federal immigration enforcement, ranging from restrictions on immigration-status inquiries to prohibitions on the use of state resources for federal enforcement [4]. As a result, Los Angeles has long positioned itself in opposition to federal immigration enforcement, shaping the strong public backlash against ICE raids.

Protests intensified on Sunday, 8 June, following President Trump's deployment of National Guard troops, justified as a response to a purported "form of rebellion" [38]. The legality and proportionality of this action were widely contested. Governor Gavin Newsom described the deployment as "a serious breach of state sovereignty" that inflamed tensions and misallocated resources (Newsom, Twitter/X). While local police were often portrayed as relatively restrained, National Guard troops – primarily stationed at federal buildings – were reported to have used force against protesters [22, 58].

Media and social media coverage reflected sharply divergent framings of the protests. Although there was broad agreement that they were sparked by ICE raids, right-leaning outlets characterized them as "riots," emphasizing violence and destruction, while supporters framed them as "demonstrations" or "community mobilization" and as legitimate responses to unconstitutional raids. Thus, even prior to incidents involving Waymo vehicles, the broader protest context was already deeply politicized and contested.

5.1.1 Vandalism of Waymo: Intended or Incidental? Waymo began as Google's Self-Driving Car Project in 2009 and was rebranded in 2016 under Google's parent company, Alphabet. Its ride-hailing service, known today as Waymo, launched in Phoenix in 2020. By 2024, the company had expanded to San Francisco, Los Angeles, and Austin, where its autonomous Jaguars became a familiar sight.

Though embraced by some with enthusiasm, Waymo's launch and expansions have also met recurring controversies over safety, regulation, and local control. Communities, including San Mateo County and Los Angeles, raised concerns about the company's autonomous vehicles operating without sufficient local oversight, citing prior incidents such as a software recall in Phoenix after crashes with a towed truck [60] and cases in L.A., where Waymo cars failed to respond correctly to traffic officers [23]. Critics have argued that local residents are being treated as "Big Tech's guinea pigs" [54], with some local officials pushing for legislation to increase municipal authority over autonomous vehicle deployment.

Prior to the June 2025 incidents, Waymo vehicles had already been targets of vandalism, particularly in San Francisco. In 2024, incidents included tire slashing across multiple robotaxis [49] and the burning of a Waymo vehicle during a crowd confrontation in Chinatown [28]. Alongside such attacks, organized grassroots resistance has challenged Waymo's presence through non-destructive tactics, such as the Safe Street Rebel group's use of "coning" to disrupt vehicles in protest of accountability, labor, and surveillance concerns [47]. Although better described as sabotage than vandalism, these actions situate the June 2025 events within a broader, politicized contestation over urban space rather than as isolated acts.

Returning to the events of June 8, media reports indicate that at least five Waymo vehicles were destroyed [44]. Protesters were observed surrounding the vehicles, tearing off doors, smashing windows, and setting the interiors on fire. No individual or group claimed responsibility or articulated specific motives. However, the presence of symbolic elements offers insight into the event's broader meanings. Images show vehicles marked with anti-ICE graffiti, directly linking the destruction to the surrounding protests. Other symbols, such as a masked protester standing atop a vehicle holding a Palestinian flag, suggest that the event carried multiple, overlapping meanings that were interpreted and enacted differently by those present.

5.1.2 Discursive Framings of Waymo and Its Vandalizers. Commentators rarely treated Waymo vehicles as social actors or entities warranting empathy; when they did, it was in a joking or sarcastic manner, drawing on popular-culture tropes of robot and AI uprisings.

In our analysis, we identified three commonly appearing interrelated interpretative frames. First, many described the cars as *private property*, condemning the vandalism ("I'm pretty sure setting private cars and damaging property aren't considered free speech" (YouTube). Others emphasized that the vehicles were *corporate property*, belonging to, and symbolizing, a large company rather than an individual. Within this frame, some commentators downplayed the harm, noting that damage to corporate-owned vehicles

is less consequential than to privately owned cars. Others, relying on this frame, highlighted how the attacks are in their view a form of resistance to tech-oligarchy, and hence are strategically understandable, or even morally justified: "These are... Not even taxi cabs because if you mess up a taxi cab that's negatively effecting a small business. It's more similar to burning a police car because you know they have good insurance. It's probably the most ethical kind of property destruction" (YouTube).

A third common frame characterized the vehicles as *surveillance tools* and instruments of state or law enforcement. Within this frame, two contrasting interpretations emerged. The first framed the destruction of the vehicles as a tactical, rather than senseless, action against surveillance. Supporting this logic, the cars' driverless design meant they could be vandalized without putting anyone at physical risk ("Technically nobody got hurt as the cars had no driver" (TikTok)). The second interpretation, common among critics of the protests, emphasized the potential role the car could play in law enforcement, suggesting that the data collected through Waymo cameras could be used to identify and prosecute the vandals.

Additional interpretations focusing on what the vandalism signified and against whom it was directed emphasized concerns about labor displacement, framing Waymo as a *threat to human drivers' livelihoods*. Others misattributed ownership, mistakenly believing the cars belonged to the police, and thus interpreted the attacks as tactical resistance against law enforcement.

These overlapping, and sometimes contradictory, interpretations to us suggested how the same event was constructed alternately as criminality, political protest, anti-capital action, or tactical resistance, each carrying distinct implications for understanding and responding to the acts.

Amid these divergent interpretations, Waymo representatives distanced the company from the events, framing the vandalism as accidental rather than targeted: "[the vehicles] were not intentionally targeted, but rather happened to be present during the protests" [12], noting that other objects, such as scooters, were also set on fire. Some online commentators echoed this view, expressing confusion and suggesting the incidents were coincidental: "Waymo has nothing to do with this. People are just angry" (TikTok).

Much like the contested interpretations of what Waymo vehicles represent, the actors involved in the vandalism – and the acts themselves – were heavily debated across news and social media. Many commentators characterized the vandalism as either, at best, foolish or stupid, or at worst, counterproductive. This critique was common across political lines: right-wing commentators emphasized that the protests were far from "peaceful," while left-leaning or protest-supportive voices highlighted that such actions could be used to discredit the broader movement. A more extreme frame involved the othering of those committing the acts, with commentators labeling them as "non-Americans" or, more derogatorily, as "third-world barbarians" or "Mexicans," suggesting that deportation was an appropriate response and added to the legitimacy of the ICE raids. In contrast, other commentaries framed the actions as a legitimate response by citizens who have exhausted conventional channels to be heard. In this reading, vandals are not criminals but lawful actors exercising their First Amendment rights in the face of systemic oppression. This perspective resonates with Ward's reflections on urban planning, where planners, through negligence or intent, can also act as vandals who are responsible for deterioration of urban infrastructure, leaving residents with little choice but to resist [63]. In this context, the federal authorities themselves are represented as the ultimate vandals, whose illegal or harmful actions leave citizens no recourse other than extreme measures.

Taken together, examining the socio-political context, mapping the actors involved, and considering their differing ideological positions helps explain why the burning of the Waymo vehicles in June 2025 became such a highly publicized and contested event. Even though the event was highly contested and disputed, the symbolic elements present during the protests and their association with broader anti-ICE and anti-surveillance sentiments suggest that the burning of the Waymo cars can be understood as a form of political expression, even if the precise targets and intentions remain ambiguous.

5.2 Case Study 2: Knightscope K5, San Francisco SPCA and the anti-homeless robot scandal

In November 2017, a Knightscope K5 security robot, which had been deployed outside the San Francisco branch of the Society for the Prevention of Cruelty to Animals (SPCA), was the subject of an attack that saw it covered, knocked over, and have barbecue sauce thrown on its sensors. It was reported, although it was unconfirmed, that the same robot had been smeared with excrement. The robot itself had been deployed in order to patrol the neighborhood surrounding the SPCA buildings, seemingly with the goal of avoiding crime, but also in a way that appeared to be seeking to target a local homeless people's encampment.

As with the Waymo case, both the vandalism and the accompanying public commentary must be situated within San Francisco's broader socio-economic context. The city has become emblematic of stark inequality driven by platform capitalism and the concentration of Big Tech firms, which has attracted high-income workers, driven up housing costs, and exacerbated homelessness [55, 66]. In 2017, 7,499 people were recorded as homeless in San Francisco [42]. The robot's deployment near the SPCA was thus embedded in an already politicized landscape shaped by housing insecurity, homelessness, and contested practices of policing.

San Francisco's homelessness and gentrification crises are thus crucial to understanding the presence of a homeless encampment and the subsequent deployment of the robot. Public concern over homelessness, housing shortages, and the policing of unhoused people therefore shaped both the robot's deployment and the online framings of its vandalism. The incident must therefore be understood within an already highly politicized urban context.

5.2.1 *Attacking Knightscope K5: mindless violence or self-protection?*

Knightscope K5 was widely reported in the press as an anti-homeless robot, deployed to patrol outside the SPCA's building. The proclaimed function of the robot was to monitor the local area and ensure that crime would be deterred. In the words of one of the spokespeople for the charity, it was intended to "improve the security around our campus and to create a safe atmosphere for staff, volunteers, clients and animals" [27]. The S.F. SPCA's president, Jennifer Scarlett, was reported as saying that the reason for the use of the robot was specifically to tackle a nearby homeless encampment. In her [20] reported words:"We weren't able to use the

sidewalks at all when there's needles and tents and bikes, so from a walking standpoint I find the robot much easier to navigate than an encampment."

The use of the patrol robot was, however, widely seen by public commentators as an insensitive and intrusive way of dealing with a serious social problem. It was viewed as targeting (rather than helping) victims of San Francisco's homelessness problem. This broadly negative sentiment, which according to reports was typically also held by the local homeless people who encountered (and were policed by) the robot, grew in intensity once the purpose of the robot's patrolling was reported in the local news. This saw headlines such as "Robots are being used to deter homeless people from setting up camp in San Francisco" [46]. In response, the local community and local authorities both began to target ongoing criticism at the SPCA on the grounds that it was using robotic technology to specifically target homeless people.

The city authorities ordered the SPCA to keep the robot off the public sidewalk or otherwise face a fine [20]. At the same time, reports circulated of (anonymous) members of the public beginning to physically attack the robot. As a result of this furore, SF SPCA was quickly forced to end its use of the Knightscope K5, with SPCA president Scarlett acknowledging that the local opposition to its use – and specifically the fact that it had been used to target homeless people – had resulted in the need to cease the robot patrol. In Scarlett's words: "Effective immediately, the San Francisco SPCA has suspended its security robot pilot program, [...] We piloted the robot program in an effort to improve the security around our campus and to create a safe atmosphere for staff, volunteers, clients and animals. Clearly, it backfired." Seeking to amend the initial explanation she had given for the use of the robot, Scarlett also stated that "The SF SPCA was exploring the use of a robot to prevent additional burglaries at our facility and to deter other crimes that frequently occur on our campus — like car break-ins, harassment, vandalism, and graffiti – not to disrupt homeless people, [...]. Reflecting the hostility that SPCA had suffered as a result of the episode, she concluded, "We regret that our words were ill-chosen. They did not properly convey the pilot program's intent and they inaccurately reflected our values."

At the same time, Knightscope sought to defend the robot's use, pointing especially to reduced crime that had been observed as a result of its patrolling: "The SPCA has the right to protect its property, employees and visitors, and Knightscope is dedicated to helping them achieve this goal. The SPCA has reported fewer car break-ins and overall improved safety and quality of the surrounding area" [16].

At the end of the episode, and having withdrawn the use of the Knightscope K5, SPCA explicitly acknowledged the threat (and two incidents) of vandalism as a factor in their decision: "Since this story has gone viral, we've received hundreds of messages inciting violence and vandalism against our facility and encouraging people to take retribution. In addition, we've already experienced two acts of vandalism on our campus" [16].

5.2.2 Framing Knightscope K5, and its vandalization. The introduction, and the withdrawal, of Knightscope K5 saw considerable online commentary. This included hundreds of online comments on Twitter/X, as well as responses to YouTube videos that reported on the story. A number of news reports also covered the story, with the Guardian reporting on interviews it had conducted directly with affected homeless people [37]. Most of this commentary tended to focus on anger towards what was widely considered to be an insensitive approach to dealing with the problem of homelessness.

In terms of the robot itself, two main ways of framing its purpose can be discerned. As a first, predominant frame, we might label the *useless, uncaring, anti-homeless robot* framing. This depicted the robot as a largely ineffective method of dealing with homelessness: "Unless it's armed – which it isn't – people will simply damage them" (YouTube). In addition, the use of the robot was widely criticised for failing to take into account the broader social context that created the homelessness: "This doesn't solve the problem – if there was housing for the homeless then there wouldn't need to be an irritating robot" (YouTube). Those who articulated this frame highlighted the irony and insensitivity of spending money on robots rather than on actually helping homeless people: "They pay for the robot but they won't give homeless people food for essentials to live" (Twitter/X). In making this observation, many noted how this was indicative of the insensitive and uncaring nature of life in San Francisco: "This is what I don't like about San Francisco, it's so uncaring to so many homeless people" (Twitter/X). One commentator also noted how this was more widely indicative of the problems of contemporary waged society: "Cheap robot security guards are less to hire than humans, and replacing them as a result" (Twitter/X).

This frame also seems to align with what we know of the motives of the homeless people themselves who interacted with the robot. For instance, the Guardian reported homeless people who were affected by the robot stating "We called it the anti-homeless robot [...] I guess that was the reason for the robot" [37]. Another homeless person was reported as saying, "That SPCA robot was the bane of our existence [...] It was driving us crazy [...] We started feeling like this thing was surveilling us for the police [...] That's officially invasion of privacy. That's uncool."

A second, less common, frame, but nevertheless one that was consistently articulated by a minority of commentators might be labelled the *effective deterrent (thwarted)* frame. This depicted the robot itself as an effective deterrent: "It stops encampments becoming excessive and keeps the city clean" (YouTube). At the same time, critics of the robot were blamed for undermining its effectiveness and ultimately resulting in its withdrawal by SPCA: "SPCA was the victim of those who want to protect the homeless rather than stop crime."

The vandalization of Knightscope K5 was most often depicted using what we might term a *vengeance imagined* framing. Whilst most commentators did not refer to the single act of vandalism that the robot actually received, nevertheless there were a large number of comments that repeatedly *imagined* inflicting violence on the robot: "I really want to destroy it" (Twitter/X). When a reason was expressed for wishing to harm the robot, this was invariably on the grounds that it was considered a negative intrusion into the lives of the homeless: "If I was homeless I'd destroy it" (Twitter/X).

Finally, reflecting a similar sentiment to those who adopted the *effective deterrent (thwarted)* frame, those commentators who were more sympathetic to the purpose of the robot were also (predictably)

more negative in sentiment regarding its vandalization: "The homeless who don't like it should move on or get a job" (YouTube). This was a much less frequently articulated frame, but it could nevertheless be observed. We might term this the *hate the homeless, hands off the robots* framing.

As we can see, both the deployment of Knightscope K5 outside the San Francisco SPCA, and its vandalization (both real and imagined), were in each instance the subject of alternative framings. The robot itself divided commentators over whether it should be considered either a *useless, uncaring, anti-homeless robot* or, alternatively, an *effective deterrent (thwarted)*. Whilst its vandalization did not receive a large amount of commentary, nevertheless the *prospect* of it being vandalized was frequently discussed. This saw an overwhelming majority adopt a *vengeance imagined* framing, which co-existed with a smaller *hate the homeless, hands off the robots* framing. In each instance, we see clear differences in terms of how the robot and its interaction with humans were framed, reflecting the broader socio-economic context and especially the position taken by the commentator regarding the nature of homelessness and the wider housing crisis of San Francisco within which this homelessness existed.

5.3 Case Study 3: The Destruction of hitchBOT in Philadelphia

A team of researchers led by Frauke Zeller and David Smith created hitchBOT in 2013 [53, 68]. The (humanoid) robot was intended to travel across public spaces by soliciting rides from humans. hitchBOT's creators framed the robot as both an artwork and a social experiment that inverted the common HRI question of whether humans can trust robots into whether robots can trust humans [53, 68]. The robot was constructed to appear "approachable" and "non-threatening": a child-sized construction of everyday objects, including a beer bucket torso and pool noodle limbs [21, 50]. It contained a GPS, a tablet computer, and a conversational interface scripted to produce "cheeky" small talk [53].

Crucially, the robot was designed without any capacity for movement, making it wholly dependent on strangers to pick it up, seat it, recharge it, and carry it onward [61]. The design inscribed a particular social image that extended into its online persona, curated through a website and social media accounts with tens of thousands of followers, where "first-person" updates, selfies, and travel commentary reinforced an anthropomorphic reading [25, 26]. This persona was further scaffolded through a quest narrative and a prewritten bucket list, positioning the machine as a quasi-social actor on a symbolic journey [11].

Before arriving in the United States, hitchBOT was taken across Canada, later transported through Germany and the Netherlands, and gained a lot of media attention and a dedicated following on social media [5, 21]. In July 2015, it was placed on the U.S. East Coast in Marblehead, Massachusetts, with San Francisco designated as its target destination [2, 61]. After being transported by two YouTubers who shared the ride on their own Twitter/X accounts, on early August 1, 2015 hitchBOT was found dismantled in Philadelphia. Some reports described it as hollowed out, prompting speculation about an acquisitive motive [6, 51]. However, no responsible person was ever identified, no police report was filed, and thus no official investigation could clarify the reasons for its destruction [31]. Photographs of the remaining parts circulated widely on social media, accompanied by a hoaxed surveillance video of the alleged incident created by the same YouTuber who last transported it [52], amplifying attention in combination with the project's already established media presence. The event quickly became the centre of a broader media spectacle that intertwined with local and national interpretations of Philadelphia as a social and moral [52, 62] landscape.

Commentaries consistently narrated the destruction not merely as vandalism of material property but through language that framed it as the harm of a living being [51] thus aligning with the common interpretations of vandalism in HRI. Headlines described hitchBOT as "murdered," "beheaded," or having "met its demise," often invoking the "life" of the artifact and drawing on kinship metaphors such as "parents," "family," and "friends." Innocence and victimhood were further underscored by repeated references to its "kid-sized" stature, its "age" of one or two years, and its popularity with children.

With hitchBOT already framed as a quasi-social actor embedded in online communities and affective discourse [68], the incident generated a second discursive stream: *the characterization of place*. Commentaries contrasted Philadelphia, and by extension the United States, with the countries where hitchBOT had previously been taken, casting the city as inhospitable and even emblematic of national moral failure [18, 52, 62]. The irony of the "City of Brotherly Love" became a recurring motif [51], at times carrying racist undertones "The trouble with the City of Brotherly Love is that it's got a few too many brothers." (Reddit), and was mobilized in headlines as well as social media jokes and memes to suggest that nothing "nice" could survive in Philadelphia [59]. Local communities attempted to resist this narrative by initiating efforts to rebuild or repair hitchBOT [31], while the researchers themselves stressed that the incident could have happened anywhere and was always an inherent risk of their experiment [18].

Yet, while drawing on a superficial symbolism of place, most interpretations left Philadelphia's socio-political context not considered. The city's challenges of poverty, homelessness, and socio-economic precarity [17, 67] did not feature in dominant accounts of the incident. One Reddit user pointed them out: "Having spent time in Philly, that tracks. There are quite a few unhoused people, and collecting scrap is a primary way many earn money" (Reddit).

In contrast, the social worlds into which hitchBOT was most visibly integrated, like wedding parties, family road trips, or public events, as well as its online persona and design aesthetics, inscribed a normative, middle-class sociality. The destruction, although officially unexplained, was largely narrated through a discourse of inhumanity, casting the unknown actor, and by extension Philadelphia residents, as cruel and unempathic. This framing left aside other possible contextualizations rooted in social need or divergent priorities. In this way, the incident became a discursive foil through which broader social inequalities and cultural stereotypes were projected and re-solidified.

To conclude, the destruction of hitchBOT to us suggested how even seemingly malicious vandalism can still be rendered a socially and politically meaningful act [10]. The dominant framing of the incident as cruelty or moral failure reveals less about the unknown

perpetrator than it does about the cultural work involved in defining who and what counts as a legitimate recipient of care and protection. An anthropomorphized, media-persona of a robot supposedly does, while a potentially marginalized person whose act of destruction may have been tied to material need or survival does not.

6 Discussion

Looking across our case studies, we find that instances of robot vandalism can be understood as ideologically significant along three interrelated dimensions. First, examining the acts themselves in context, particularly in the Waymo and Knightscope K5 cases, reveals evidence of ideological intent [10]. The symbolism embedded in these actions, alongside their alignment with broader socio-political concerns, suggests these were not "senseless" (meaning deprived) gestures of aggression. Rather, the acts can be read as material and symbolic interventions expressing resistance to politically charged events, grounded in political belief systems (e.g., anti-ICE, anti-surveillance). Importantly, these beliefs are not necessarily singular: multiple ideological positions can coexist within a single act, and at times they may even be in tension or contradiction, reflecting the complex socio-political meanings participants themselves attach to their actions.

Second, following Fairclough's view [15] of discourse as both shaped by and shaping social practices, the destruction of robots becomes political through the discursive struggle over their meaning. In the Waymo case, the burning of vehicles was widely publicized and leveraged as a political talking point across partisan lines. Competing interpretations (Was it a riot or a protest? A criminal act or an act of civic resistance?) served to assert particular versions of social reality and legitimize specific understandings of the state, technology, and civic order. Similarly, in the Knightscope K5 case, discursive framings of the vandalism depended on whether homelessness was seen as the result of individual moral failure or broader socio-economic conditions, revealing underlying ideological assumptions about causality and responsibility.

Third, the politics of the artefacts themselves must be considered. As we highlighted throughout the paper, robots are never just material or social actors; they are socio-technical and socio-political artifacts [65]. Related to that, the hitchBOT case allowed us to explore how even apparently non-ideological or, on the surface, senseless destructive acts acquire political significance when we examine the assumptions and values embedded in a robot's design and narratives. Conceived as highly anthropomorphized hitchhiking companion, hitchBOT reflected middle-class, Western cosmopolitan ideals. To us, its destruction in Philadelphia, a city marked by economic precarity and struggles, highlighted how the political significance of its destruction can emerge indirectly, through the friction between the expectations embedded in a robot's design and the lived realities of the communities that encounter it.

With regards to the implications for the HRI community, one might first ask: What does it matter how we label robot vandalism when the outcome is the same? We argue that classifying such acts – either as malicious destruction, criminality, or political resistance – is not merely a theoretical exercise. Rather, it directly shapes how scholars and designers understand and respond to these events. Interpreting all vandalism as senseless because it violates our normative expectations of property and order risks obscuring and neglecting the broader systemic and structural forms of violence that underlie these outbursts [69]. Consequently, mitigation strategies that focus solely on system-level design or public education, even if effective in narrow terms, risk to remain short-term solutions that do not contribute to the social sustainability of the systems we design.

Assuming we agree on the importance of understanding of the structural and systemic issues for how we conceptually approach and design, how can HRI researchers in fact grapple with the socio-political roots of vandalism? Our hope is that the work presented here exemplified how unpacking ideological underpinning of vandalism requires methodological approaches that exceed current dominant methodological practices in HRI that focus on the micro-level of interactions. Social-science-informed methods – be it critical discourse analysis, frame analysis, or other approaches from critical anthropology – are crucial when we want to examine how human-robot interactions acquire meaning and political valence within broader structures of ideology, representation, and material power. We thus call for further incorporation of such methods, and respective epistemologies, into the HRI toolkit.

Lastly, recognizing the structural and political dimensions of vandalism to us gestures toward the importance of community-oriented and justice-oriented design [9]. Importantly, we distinguish community-oriented design from those practices of participatory design that engage pre-selected participants to validate and, in best case, inform, for the most part predetermined decisions [1, 33]. In contrast, following Suchman, we understand community-centered design as a practice of located accountability, embedded from the outset as a political project [56, 57]. This approach requires reflexivity on the part of researchers and designers, including critical engagement with their own political positionality and alliances, careful consideration of who the system is for, and attention to the often contested political realities of the community members involved. Importantly, in some cases, this may lead to the conclusion that a system should not be deployed at all. Ultimately, for us, community-oriented design highlights that interactions with robots are shaped less, or not exclusively, by technical features than by the social functions they perform and the power relations they embody.

7 Conclusion

Our study offered three key contributions to HRI. First, it reframed robot vandalism as a socially and politically embedded phenomenon rather than an isolated interactional breakdown. Second, it underscored the importance of incorporating social-science-informed methods that attend to the structural and political dimensions of human–robot encounters. Third, it called for community- and justice-oriented design approaches that engage directly with the social realities, conflicts, and power relations shaping robots' presence in public life.

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